

- [Skip to main navigation](#)
- [Skip to search](#)
- [Skip to main content](#)

[Coventry University Home](#) [Coventry University Logo](#)

- [Help & FAQ](#)
- [Home](#)
- [Organisational Units](#)
- [Profiles](#)
- [Research output](#)
- [Projects](#)
- [Coventry awarded Student theses](#)
- [Activities](#)
- [Prizes](#)
- [Datasets](#)
- [Press/Media](#)
- [Equipment](#)

Search by expertise, name or affiliation

Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial

Neda Haghighat, Behzad Nazarian, Cain C T Clark, Ladan Aghakhani

- [Shiraz University of Medical Sciences](#)
- [Lorestan University of Medical Sciences](#)

Research output: Contribution to journal › Article › peer-review

- [Overview](#)
- [Fingerprint](#)

Abstract

Background & Aims: Folic acid is a widely used supplement, with numerous purported health benefits. The purpose of this study was to evaluate the effect of folic acid on body weight (BW) and/or body mass index (BMI). **Methods:** This systematic review and meta-analysis of 11 randomized controlled trials (RCTs) evaluated the effects of folic acid supplementation

BW or BMI in 514 participants. A systematic search, current as of March 2021, was performed in PubMed/Medline, Scopus, Web of Science, and Embase using relevant keywords to identify eligible studies. A fixed-effects model was used to estimate the weighted mean difference (WMD) and 95% confidence intervals (CIs).

Results: 11 RCTs were included in this meta-analysis. The pooled analysis revealed that folic acid supplementation did not affect BW (WMD: 0.16 kg; 95% CI: -0.77 to 1.10; $p = 0.727$, $I^2 = 0.0\%$, $p = 1.00$) and BMI (WMD: 0.05 kg/m²; 95% CI: -0.08 to 0.19; $p = 0.447$, $I^2 = 0.0\%$, $p = 0.666$).

Conclusions: Folic acid supplementation did not significantly impact body weight or BMI. Future large-scale RCTs, on various dosages and durations of intervention, are needed to elucidate the effects more clearly.

Original language	English
Pages (from-to)	642-650
Number of pages	9
Journal	Current Nutrition and Food Science
Volume	19
Issue number	6
Early online date	23 Nov 2022

DOIs • <https://doi.org/10.2174/1570159x21666221017140005>

Publication status Published - Jul 2023

Keywords

- folic acid
- folate
- body weight
- body mass index
- obesity
- meta-analysis

UN SDGs

This output contributes to the following UN [Sustainable Development Goals \(SDGs\)](#)

- SDG 3 - Good Health and Well-being

Access to Document

- [10.2174/1570159x21666221017140005](https://doi.org/10.2174/1570159x21666221017140005)

Other files and links

- [Link to publication in Scopus](#)

Fingerprint

Dive into the research topics of 'Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial'. Together they form a unique fingerprint.

-

Folic Acid
Nursing and Health Professions
100%

-

Body Weight
Nursing and Health Professions
83%

-

Body Mass
Nursing and Health Professions
83%

-

Supplementation
Nursing and Health Professions
66%

-

Randomized Controlled Trial
Pharmacology, Toxicology and Pharmaceutical Science
66%

- 

Confidence Interval
Nursing and Health Professions
50%

- 

Meta Analysis
Nursing and Health Professions
50%

- 

Systematic Review
Nursing and Health Professions
33%

[View full fingerprint](#)

Cite this

- APA
- Standard
- Harvard
- Vancouver
- Author
- BIBTEX
- RIS

Haghighat, N., Nazarian, B., Clark, C. C. T., & Aghakhani, L. (2023). [Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial](#). [Current Nutrition and Food Science](#), 19(6), 642-650. <https://doi.org/10.2174/1570159x21666221017140005>
[Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial](#). / Haghighat, Neda; Nazarian, Behzad; Clark, Cain C T et al.

In: [Current Nutrition and Food Science](#), Vol. 19, No. 6, 07.2023, p. 642-650.

Research output: Contribution to journal › Article › peer-review

Haghighat, N, Nazarian, B, Clark, CCT & Aghakhani, L 2023, 'Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial', [Current Nutrition and Food Science](#), vol. 19, no. 6, pp. 642-650. <https://doi.org/10.2174/1570159x21666221017140005>

Haghighat N, Nazarian B, Clark CCT, Aghakhani L. [Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial](#). *Current Nutrition and Food Science*. 2023 Jul;19(6):642-650. Epub 2022 Nov 23. doi: 10.2174/1570159x21666221017140005

Haghighat, Neda ; Nazarian, Behzad ; Clark, Cain C T et al. / [Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial](#).

In: *Current Nutrition and Food Science*. 2023 ; Vol. 19, No. 6. pp. 642-650.

@article{c94892e9f0184475a4c764753b202681,

title = "Does folic acid supplementation affect body weight or body mass index? A systematic review and meta-analysis of randomized controlled trial",

abstract = "Background & Aims: Folic acid is a widely used supplement, with numerous purported health benefits. The purpose of this study was to evaluate the effect of folic acid on body weight (BW) and/or body mass index (BMI). Methods: This systematic review and meta-analysis of 11 randomized controlled trials (RCTs) evaluated the effects of folic acid supplementation BW or BMI in 514 participants. A systematic search, current as of March 2021, was performed in PubMed/Medline, Scopus, Web of Science, and Embase using relevant keywords to identify eligible studies. A fixed-effects model was used to estimate the weighted mean difference (WMD) and 95% confidence intervals (CIs).Results: 11 RCTs were included in this meta-analysis. The pooled analysis revealed that folic acid supplementation did not affect BW (WMD: 0.16 kg; 95% CI: -0.77 to 1.10; p = 0.727, I²= 0.0%, p = 1.00) and BMI (WMD: 0.05 kg/m²; 95% CI: -0.08 to 0.19; p = 0.447, I²= 0.0%, p = 0.666). Conclusions: Folic acid supplementation did not significantly impact body weight or BMI. Future large-scale RCTs, on various dosages and durations of intervention, are needed to elucidate the effects more clearly.",

keywords = "folic acid, folate, body weight, body mass index, obesity, meta-analysis",

author = "Neda Haghighat and Behzad Nazarian and Clark, {Cain C T} and Ladan Aghakhani",

year = "2023",

month = jul,

doi = "10.2174/1570159x21666221017140005",

language = "English",

volume = "19",

pages = "642--650",

journal = "Current Nutrition and Food Science",

issn = "1573-4013",

publisher = "Bentham Science Publishers Ltd.",

number = "6",

}

TY - JOUR

T1 - Does folic acid supplementation affect body weight or body mass index?
A systematic review and meta-analysis of randomized controlled trial

AU - Haghighat, Neda

AU - Nazarian, Behzad

AU - Clark, Cain C T

AU - Aghakhani, Ladan

PY - 2023/7

Y1 - 2023/7

N2 - Background & Aims: Folic acid is a widely used supplement, with numerous purported health benefits. The purpose of this study was to evaluate the effect of folic acid on body weight (BW) and/or body mass index (BMI). Methods: This systematic review and meta-analysis of 11 randomized controlled trials (RCTs) evaluated the effects of folic acid supplementation BW or BMI in 514 participants. A systematic search, current as of March 2021, was performed in PubMed/Medline, Scopus, Web of Science, and Embase using relevant keywords to identify eligible studies. A fixed-effects model was used to estimate the weighted mean difference (WMD) and 95% confidence intervals (CIs). Results: 11 RCTs were included in this meta-analysis. The pooled analysis revealed that folic acid supplementation did not affect BW (WMD: 0.16 kg; 95% CI: -0.77 to 1.10; $p = 0.727$, $I^2 = 0.0\%$, $p = 1.00$) and BMI (WMD: 0.05 kg/m²; 95% CI: -0.08 to 0.19; $p = 0.447$, $I^2 = 0.0\%$, $p = 0.666$). Conclusions: Folic acid supplementation did not significantly impact body weight or BMI. Future large-scale RCTs, on various dosages and durations of intervention, are needed to elucidate the effects more clearly.

AB - Background & Aims: Folic acid is a widely used supplement, with numerous purported health benefits. The purpose of this study was to evaluate the effect of folic acid on body weight (BW) and/or body mass index (BMI). Methods: This systematic review and meta-analysis of 11 randomized controlled trials (RCTs) evaluated the effects of folic acid supplementation BW or BMI in 514 participants. A systematic search, current as of March 2021, was performed in PubMed/Medline, Scopus, Web of Science, and Embase using relevant keywords to identify eligible studies. A fixed-effects model was used to estimate the weighted mean difference (WMD) and 95% confidence intervals (CIs). Results: 11 RCTs were included in this meta-analysis. The pooled analysis revealed that folic acid supplementation did not affect BW (WMD: 0.16 kg; 95% CI: -0.77 to 1.10; $p = 0.727$, $I^2 = 0.0\%$, $p = 1.00$) and BMI (WMD: 0.05 kg/m²; 95% CI: -0.08 to 0.19; $p = 0.447$, $I^2 = 0.0\%$, $p = 0.666$). Conclusions: Folic acid supplementation did not significantly impact body weight or BMI. Future large-scale RCTs, on various dosages and durations of intervention, are needed to elucidate the effects more clearly.

KW - folic acid

KW - folate

KW - body weight

KW - body mass index

KW - obesity

KW - meta-analysis

UR - <http://www.scopus.com/inward/record.url?scp=85160809999&partnerID=8YFLogxK>

U2 - 10.2174/1570159x21666221017140005

DO - 10.2174/1570159x21666221017140005

M3 - Article

SN - 1573-4013

VL - 19

SP - 642

EP - 650

JO - Current Nutrition and Food Science

JF - Current Nutrition and Food Science

IS - 6

ER -



Powered by [Pure](#), [Scopus](#) & [Elsevier Fingerprint Engine™](#)

All content on this site: Copyright © 2024 [Elsevier B.V.](#) or its licensors and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the Creative Commons licensing terms apply

We use cookies to help provide and enhance our service and tailor content. By continuing you agree to the [use of cookies](#)



[Log in to Pure](#)

[Coventry University data protection policy](#)

[About web accessibility](#)

[Report vulnerability](#)

[Contact us](#)